R Code

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# PRACTICAL 3: STUDY OF STATIONARITY, ACF AND PACF
#             USING R

# Objective:
# To study the characteristics of a time series using real-world
# data and to examine its autocorrelation function (ACF) and
# partial autocorrelation function (PACF) using R.
#
# The practical will help us understand:
# 1. Time series data and its time structure
# 2. Visual identification of stationarity
# 3. The concept of lag
# 4. Autocorrelation function (ACF)
# 5. Partial autocorrelation function (PACF)
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# 1. INTRODUCTION TO THE DATASET
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# We use the inbuilt R dataset "Nile".
#
# The Nile dataset contains annual measurements of the flow
# of the River Nile at Ashwan for the years 1871 to 1970.
#
# Since the observations are recorded annually, the frequency
# of the time series is 1.

data(Nile)

# Display the complete dataset
Nile


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# 2. BASIC INFORMATION ABOUT THE TIME SERIES
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# Check the class of the object

class(Nile)

# "ts" indicates that Nile is a time-series object.


# Find the total number of observations

length(Nile)
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# Find the starting time of the series

start(Nile)


# Find the ending time of the series

end(Nile)


# Find the frequency of the series

frequency(Nile)

# Since the data are annual:
#
# frequency = 1
#
# This means that there is one observation per year.


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# 3. VISUALIZATION OF THE TIME SERIES
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# Plot the original time series

plot(Nile,
     main = "Annual Flow of the River Nile",
     xlab = "Year",
     ylab = "River Flow")


# Carefully observe the plot.
#
# 1. Is there a strong increasing or decreasing trend?
# 2. Does the series fluctuate around a relatively stable level?
# 3. Does the variability appear approximately constant?
# 4. Are there any unusual observations or changes in level?
#
# These observations help us get an initial idea about
# whether the series can be considered approximately stationary.



# 4. AUTOCORRELATION FUNCTION (ACF)
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# The autocorrelation function measures the correlation
# between observations separated by different time lags.

acf(Nile,
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    main = "Autocorrelation Function of Nile River Flow")


# Interpretation of the ACF plot:
#
# X-axis -> Lag
#
# Y-axis -> Autocorrelation
#
# Each vertical bar represents the estimated autocorrelation
# at a particular lag.
#
# The horizontal dashed lines represent approximate
# 95% confidence limits.
#
# A bar extending beyond the confidence limits suggests
# that the corresponding autocorrelation is significantly
# different from zero.


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# 5. DISPLAYING THE ACF VALUES NUMERICALLY
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# The acf() function can also be stored as an object.

acf_nile <- acf(Nile,
                plot = FALSE)

# Display the ACF object

acf_nile


# Extract the autocorrelation values

acf_nile$acf


# Extract the corresponding lag values

acf_nile$lag


# Display the first few autocorrelation values:

acf_nile$acf[1:6]


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# 6. PARTIAL AUTOCORRELATION FUNCTION (PACF)
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# PACF measures the correlation between Yt and Y(t-k)
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# after accounting for the effects of the intermediate lags.

pacf(Nile,
     main = "Partial Autocorrelation Function of Nile River Flow")


# Interpretation:
#
# X-axis -> Lag
#
# Y-axis -> Partial autocorrelation
#
# Each vertical bar represents the partial autocorrelation
# at the corresponding lag.
#
# The dashed horizontal lines represent approximate
# 95% confidence limits.


######################################################################
# 7. DISPLAYING PACF VALUES NUMERICALLY
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# Store the PACF results without plotting

pacf_nile <- pacf(Nile,
                  plot = FALSE)

# Display the PACF object

pacf_nile


# Extract PACF values

pacf_nile$acf


# Extract the corresponding lag values

pacf_nile$lag


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# 8. COMPARING ACF AND PACF
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# Plot ACF

acf(Nile,
    main = "ACF of Nile River Flow")


# Plot PACF
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pacf(Nile,
     main = "PACF of Nile River Flow")



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# 9. COMPARISON WITH A NON-STATIONARY TIME SERIES
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# In the previous practical, we studied AirPassengers.
# We now use it only to compare the behaviour of ACF and PACF.
#
# AirPassengers contains a strong trend and seasonal pattern.

data(AirPassengers)

# Plot AirPassengers

plot(AirPassengers,
     main = "AirPassengers Time Series",
     xlab = "Year",
     ylab = "Number of Passengers")


# ACF of AirPassengers

acf(AirPassengers,
    main = "ACF of AirPassengers")


# PACF of AirPassengers

pacf(AirPassengers,
     main = "PACF of AirPassengers")


# Compare these plots with the ACF and PACF of the Nile data.
#
# Think about the following:
#
# 1. Is the ACF decreasing quickly or slowly?
# 2. Are there large autocorrelations at many lags?
# 3. Is there a seasonal pattern visible in the ACF?
# 4. How does the ACF of AirPassengers differ from that
#    of the Nile series?
#
# This comparison helps us understand why stationarity is
# important before developing time-series models.



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# PRACTICE QUESTION
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# Use the inbuilt R dataset "co2".

# Perform the following analysis:
#
# (a) Load and display the co2 dataset.
#
# (b) Find:
#     - Class
#     - Length
#     - Starting time
#     - Ending time
#     - Frequency
#
# (c) Plot the original time series.
#
# (d) Examine the plot and comment on whether the series
#     appears stationary or non-stationary.
#
# (e) Plot the ACF of the co2 series.
#
# (f) Plot the PACF of the co2 series.
#
# (g) Compare the ACF and PACF plots of co2 with those
#     obtained for the Nile dataset.
#
# (h) Comment on the dependence between observations at
#     different lags.

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